The Unbalanced Dual Matching Coefficients: $\boldsymbol{\Gamma(\frac{1}{4}+ix)\,\Gamma(\frac{3}{4}-ix)}$

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1 The Kernel and Its Shift

Consider the *bilinear* pairing $\langle f, g\rangle = \int_{\mathbb{R}} f\,g\,dx$ (no conjugation) and the kernel

$$W(x) = \Gamma(\tfrac{1}{4}+ix)\,\Gamma(\tfrac{3}{4}-ix).$$

By the reflection formula $\Gamma(s)\Gamma(1-s) = \pi/\sin\pi s$ with $s = \frac{1}{4}+ix$,

$$W(x) = \frac{\pi}{\sin\pi(\frac{1}{4}+ix)}, \tag{1}$$

which is *complex-valued*, with $W(-x) = \overline{W(x)}$. In particular W is not a positive weight and no Hermitian orthogonal polynomials exist—this is genuine dual-basis territory.

The shift. The imaginary translation $y = x + \frac{i}{4}$ maps the kernel to

$$W(y - \tfrac{i}{4}) = \Gamma(\tfrac{1}{2}+iy)\,\Gamma(\tfrac{1}{2}-iy) = \left|\Gamma(\tfrac{1}{2}+iy)\right|^2,$$

the standard weight for Meixner–Pollaczek polynomials with $\lambda = \frac{1}{2}$, $\phi = \frac{\pi}{2}$. This reduces the bilinear problem to a shifted copy of a classical self-dual case.

Specifically, let P_n denote the standard $\mathrm{MP}(\frac{1}{2}, \frac{\pi}{2})$ polynomials, satisfying

$$(n+1)\,P_{n+1}(y) = 2y\,P_n(y) - n\,P_{n-1}(y), \qquad P_0 = 1, \quad P_1 = 2y,$$

with norms $h_n^{\mathrm{std}} = \int_{\mathbb{R}} P_m P_k\,|\Gamma(\frac{1}{2}+iy)|^2\,dy = \pi\,\delta_{mk}$ (all norms equal π). Then $Q_n(x) := P_n(x+\frac{i}{4})$ are the monic bilinear orthogonal polynomials for W:

$$\int_{\mathbb{R}} Q_m(x)\,Q_k(x)\,W(x)\,dx = \pi\,\delta_{mk}.$$

The monic versions $q_n = (n!/2^n)\,Q_n$ satisfy the three-term recurrence

$$q_{n+1} = \left(x + \tfrac{i}{4}\right) q_n - \left(\tfrac{n}{2}\right)^2 q_{n-1}, \qquad h_n^{\mathrm{monic}} = \pi\,\frac{(n!)^2}{4^n}. \tag{2}$$

(Verified numerically to 50 digits for $n \le 11$.)

2 Moments

The two-Gamma integral formula gives the moment generating function. With $a = \frac{1}{4}$, $b = \frac{3}{4}$,

$$\int_{\mathbb{R}} W(x)\,e^{isx}\,dx = 2\pi\,\frac{e^{3s/4}}{1+e^s}\,, \tag{3}$$

so $\mu_0 = \pi$, and the moments $\mu_n = \int x^n W\,dx$ are obtained by differentiating: $\mu_n = (-i)^n F^{(n)}(0)$ where $F(s) = 2\pi e^{3s/4}/(1+e^s)$. Even moments are real, odd moments purely imaginary (from $W(-x) = \overline{W(x)}$), and the Hankel determinants $\Delta_n = \det(\mu_{j+k})_{0\le j,k<n}$ are all real and positive (verified for $n \le 12$), confirming existence of the bilinear orthogonal polynomials.

3 The Three-Gamma Matching Integral

Split $W = w_+ \, w_-$ with $w_+ = \Gamma(\frac{1}{4} + ix)$, $w_- = \Gamma(\frac{3}{4} - ix)$. The normalised dual functions are

$$F_m = w_+ \, \widehat{P}_m, \qquad G_m = w_- \, \widehat{P}_m,$$

biorthonormal: $\int F_m \, G_k \, dx = \delta_{mk}$ (verified to 50 digits for $m, k \le 6$). The matching coefficients

$$c_{mk} = \langle F_m, \widehat{P}_k \rangle_W = \frac{J_{mk}}{\pi}, \qquad J_{mk} = \int_{\mathbb{R}} P_m\big(x + \tfrac{i}{4}\big) \, P_k\big(x + \tfrac{i}{4}\big) \, \Gamma\big(\tfrac{1}{4} + ix\big)^2 \, \Gamma\big(\tfrac{3}{4} - ix\big) \, dx,$$

are computed by the three-Gamma integral (Proposition 1 of the companion paper) with $a = b = \frac{1}{4}$, $c = \frac{3}{4}$:

$$J_{00} = 2\pi \, \Gamma(1)^2 \, U(1,1,1) = 2\pi \, e \, E_1(1) = 3.74696098492\ldots \tag{4}$$

using $U(1,1,z) = e^z \, E_1(z)$ (Tricomi's function in terms of the exponential integral).

4 The Master Generating Function

Substituting the polynomial generating function $\sum P_n(y) \, w^n = (1 - iw)^{-1/2+iy} (1 + iw)^{-1/2-iy}$ with $y = x + \frac{i}{4}$ into the matching integral and using the shifted three-Gamma formula (equation (2) of the companion paper),

$$\boxed{\sum_{m,k \ge 0} J_{mk} \, w^m u^k = \frac{2\pi \, e^{\eta} \, E_1(\eta)}{(1 - iw)(1 - iu)}, \qquad \eta = \frac{(1 + iw)(1 + iu)}{(1 - iw)(1 - iu)}.} \tag{5}$$

At $w = u = 0$, $\eta = 1$, recovering (4). Verified numerically to 15 digits against all 78 entries of the 12×12 table.

Comparison with the self-dual case

The companion paper's generating function for $W_0 = |\Gamma(\frac{1}{4} + ix)|^2$ (the self-dual MP($\frac{1}{4}$) case) was

$$\frac{2\pi^{3/2} \, e^{\zeta/2} \, K_0(\zeta/2)}{\sqrt{(1 - iw)(1 - iu)}}, \qquad \zeta = \varphi(w) \, \varphi(u),$$

with φ the Cayley transform and the modified Bessel function K_0. The structural differences are summarised:

	Self-dual ($\lambda = \frac{1}{4}$)	Unbalanced ($\lambda = \frac{1}{2}$ shifted)
Special fn	$K_0(\zeta/2)$	$E_1(\eta)$
ODE order	2 (Bessel)	1 (first-order)
Generators	2: $A = \sqrt{2e} \, K_0(\frac{1}{2})$, $B = \sqrt{2e} \, K_1(\frac{1}{2})$	1: $A = e \, E_1(1)$
Prefactor	$1/\sqrt{(1 - iw)(1 - iu)}$	$1/\big((1 - iw)(1 - iu)\big)$
Variable	$\zeta = \varphi(w)\varphi(u)$	$\eta = \frac{(1+iw)(1+iu)}{(1-iw)(1-iu)}$

Single-generator structure

The derivative algebra of $f(z) = e^z E_1(z)$ is first-order:

$$f'(z) = f(z) - \frac{1}{z},$$

so $f^{(n)}(1) \in \mathbb{Q} \cdot A \oplus \mathbb{Q}$ for all n by induction. Since the Taylor coefficients of (5) are polynomial expressions in the derivatives of f at $z = 1$, every matching coefficient lies in $\mathbb{Q}\,A \oplus \mathbb{Q}$:

$$\frac{J_{mk}}{2\pi} \in (\mathbb{Q}\,A \oplus \mathbb{Q}) \cdot i^{m+k}, \qquad A = e\,E_1(1) = 0.59634736\ldots \tag{6}$$

This contrasts with the self-dual case, where the Bessel ODE is second-order and produces *two* independent generators. The structural reason: E_1 satisfies $zE_1' + E_1 = e^{-z}/z$, a first-order inhomogeneous equation, while K_0 satisfies $z^2K_0'' + zK_0' - z^2K_0 = 0$, a second-order equation whose solution space is two-dimensional.

Parity

Since $W(-x) = \overline{W(x)}$, substituting $x \mapsto -x$ gives $J_{mk} = (-1)^{m+k}\overline{J_{mk}}$: entries with $m+k$ even are real, odd are purely imaginary. The self-dual case had the same checkerboard; the kernel's asymmetry does not break it.

5 The Coefficient Table

The table lists $J_{mk}/(2\pi)$ decomposed over the single generator $A = e\,E_1(1)$. A factor of i is suppressed for $m+k$ odd.

(m,k)	$J_{mk}/(2\pi)$	(m,k)	$J_{mk}/(2\pi)$
$(0,0)$	A	$(1,3)$	$87A - 52$
$(0,1)$	$3A - 2$	$(1,4)$	$\frac{547}{3}A - \frac{326}{3}$
$(0,2)$	$-7A + 4$	$(2,2)$	$121A - 72$
$(0,3)$	$-\frac{43}{3}A + \frac{26}{3}$	$(2,3)$	$\frac{949}{3}A - \frac{566}{3}$
$(0,4)$	$27A - 16$	$(2,4)$	$-\frac{2167}{3}A + \frac{1292}{3}$
$(0,5)$	$\frac{719}{15}A - \frac{86}{3}$	$(3,3)$	$-\frac{8101}{9}A + \frac{4832}{9}$
$(1,1)$	$-13A + 8$	$(3,4)$	$-\frac{6617}{3}A + \frac{3946}{3}$
$(1,2)$	$-37A + 22$	$(4,4)$	$\frac{51649}{9}A - \frac{30800}{9}$

All PSLQ identifications verified to 40+ digits. The matching coefficients of F_m are $c_{mk} = J_{mk}/\pi$; those of G_m are the complex conjugates.

6 The Reverse Expansion

The reverse coefficients $D_{mk} = \int \widehat{P}_m\,G_k\,dx$ involve a single Gamma factor. The one-Gamma Fourier pair gives

$$\int_{\mathbb{R}} \Gamma(\tfrac{3}{4} - ix)\,e^{isx}\,dx = 2\pi\,e^{3s/4}\,e^{-e^s},$$

and the generating function for the reverse table is

$$\sum_{m,k}\Big(\int P_m P_k\,\Gamma(\tfrac{3}{4} - ix)\,dx\Big)w^m u^k = \frac{2\pi\,e^{-1/\eta}}{(1 - iw)(1 - iu)}\,, \tag{7}$$

with η as before. At $w = u = 0$: $2\pi/e$. The single generator is now e^{-1}—the exponential integral has vanished, leaving pure exponential arithmetic. Every reverse coefficient is a Gaussian rational times $2\pi/e$, provably so from the Taylor expansion.

First entries of $R_{mk}/(2\pi/e)$, with i suppressed for $m+k$ odd:

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{4} & \frac{1}{2} & -\frac{11}{4} \\ & \frac{3}{4} & \frac{1}{4} & \frac{13}{8} & -\frac{15}{4} & -\frac{49}{8} \\ & & \frac{9}{4} & -\frac{25}{8} & \frac{1}{4} & -\frac{171}{8} \end{pmatrix}$$

7 Zeros

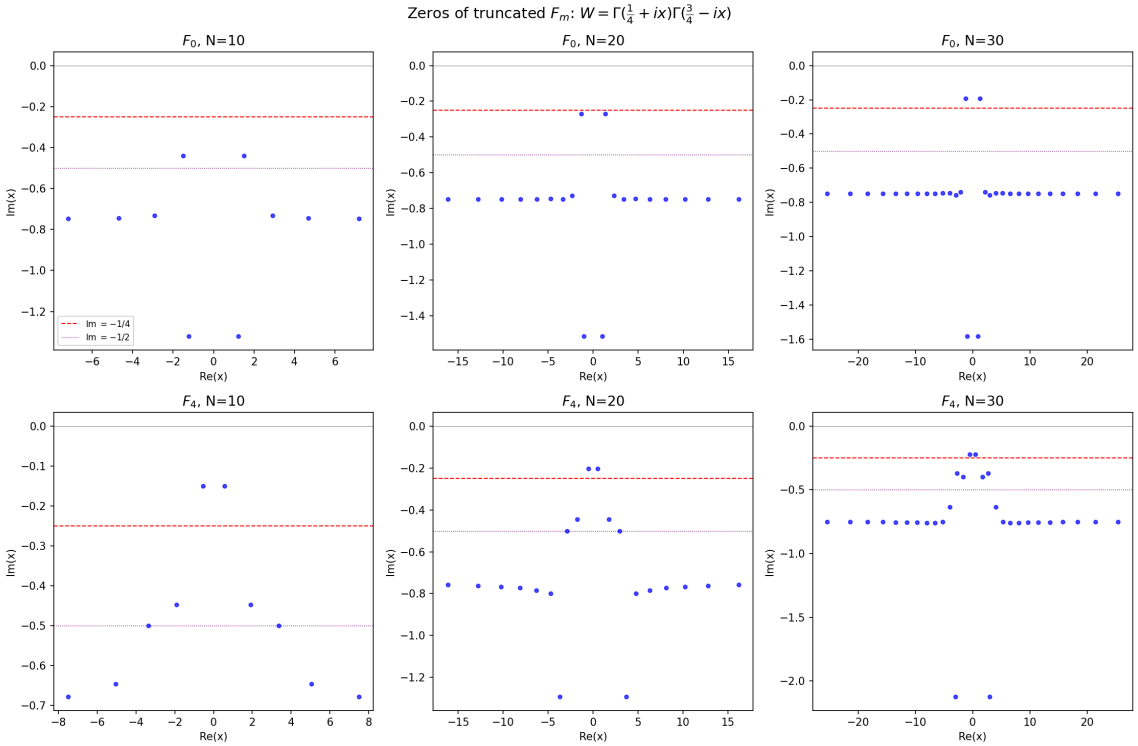


Figure 1: Zeros of the truncated dual functions $\sum_{k\le N} c_{mk}\,\widehat{P}_k$ for $m = 0$ (top) and $m = 4$ (bottom), at truncation orders $N = 10, 20, 30$. The bulk of the spurious zeros lie in a band near $\operatorname{Im} x \approx -0.7$, close to the nearest pole line $\operatorname{Im} x = -\frac{3}{4}$ of $\Gamma(\frac{3}{4} - ix)$.

In the self-dual case $W_0 = |\Gamma(\frac{1}{4} + ix)|^2$, the spurious zeros of the polynomial approximation concentrated on the line $\operatorname{Im} x = -\frac{1}{2}$, midway between the pole ladders at $\pm i(\frac{1}{4} + \mathbb{Z}_{\ge 0})$. Here the pole ladders are asymmetric:

$$w_+ = \Gamma(\tfrac{1}{4} + ix): \quad \text{poles at } x = i(\tfrac{1}{4} + n), \quad n \ge 0,$$
$$w_- = \Gamma(\tfrac{3}{4} - ix): \quad \text{poles at } x = -i(\tfrac{3}{4} + n), \quad n \ge 0.$$

The nearest pole to the real axis from *below* is at $\operatorname{Im} x = -\frac{3}{4}$, and Figure 1 shows the bulk zeros cluster near this line rather than at the midpoint $-\frac{1}{4}$. The zero-band is pulled towards the nearer w_- pole ladder, reflecting the asymmetry of the kernel.

8 Discussion

The unbalanced kernel $\Gamma(\frac{1}{4} + ix)\,\Gamma(\frac{3}{4} - ix)$ illustrates three phenomena:

1. **Reduction by shift.** Every kernel $\Gamma(\lambda + ix)\,\Gamma(1 - \lambda - ix)$ for $0 < \lambda < 1$ reduces via $y = x + i(\frac{1}{2} - \lambda)$ to the self-dual weight $|\Gamma(\frac{1}{2} + iy)|^2$, i.e. the MP($\frac{1}{2}$) case. The "unbalanced" appearance is an artifact of the real-line presentation.

2. **First-order vs. second-order.** The exponential integral E_1 replaces the Bessel function K_0 of the self-dual ($\lambda = \frac{1}{4}$) case. The derivative algebra collapses from a two-state automaton ($K_0 \leftrightarrow K_1$) to a one-state system ($A \mapsto A - 1$), halving the number of generators. The structural reason is the ODE order: K_ν satisfies a second-order Bessel equation, while E_1 satisfies a first-order equation.

3. **Zeros geometry.** The spurious zero band shifts from $\operatorname{Im} x = -\frac{1}{2}$ (self-dual) to $\operatorname{Im} x \approx -\frac{3}{4}$ (unbalanced), tracking the nearer pole ladder of the split.

The broader family $\Gamma(\lambda + ix)\,\Gamma(\mu - ix)$ with $\lambda + \mu = 1$ (reflection constraint) always reduces to MP($\frac{1}{2}$) by translation, but when $\lambda + \mu \neq 1$ the situation is genuinely different—the three-Gamma integral involves $U(\alpha, \beta, 1)$ with $\alpha \neq \frac{1}{2}$ and the Bessel/exponential-integral correspondence breaks. This will be addressed in a sequel.